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EEE 421

RF and Microwave Engineering

Waveguide and Microstrips

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Modes of Wave Propagation

A wave has both electric and magnetic fields. All transverse components of electric and magnetic fields are determined from the axial components of electric and magnetic field, in the z direction. This allows mode formations, such as TE, TM, TEM and Hybrid in microwaves. Let us have a look at the types of modes.

Types of Modes

The modes of propagation of microwaves are –

TEM (Transverse Electromagnetic Wave)

In this mode, both the electric and magnetic fields are purely transverse to the direction of propagation. There are no components in 'Z' direction.

$$E_z = 0 \text{ and } H_z = 0$$

TE (Transverse Electric Wave)

In this mode, the electric field is purely transverse to the direction of propagation, whereas the magnetic field is not.

$$E_z = 0 \text{ and } H_z \neq 0$$

TM (Transverse Magnetic Wave)

In this mode, the magnetic field is purely transverse to the direction of propagation, whereas the electric field is not.

$$E_z \neq 0 \text{ and } H_z = 0$$

HE (Hybrid Wave)

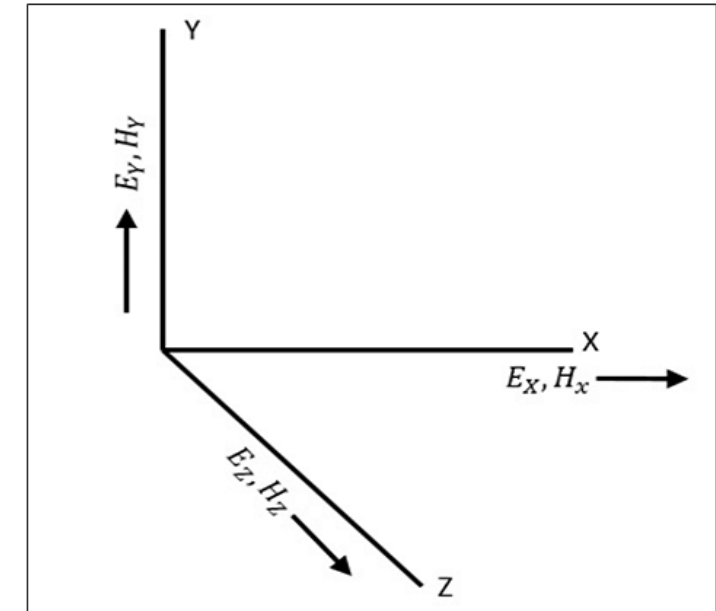
In this mode, neither the electric nor the magnetic field is purely transverse to the direction of propagation.

$$E_z \neq 0 \text{ and } H_z \neq 0$$

Multi conductor lines normally support TEM mode of propagation, as the theory of transmission lines is applicable to only those system of conductors that have a go and return path, i.e., those which can support a TEM wave.

Waveguides are single conductor lines that allow TE and TM modes but not TEM mode. Open conductor guides support Hybrid waves.

The direction of the electric and the magnetic field components along three mutually perpendicular directions x, y, and z are as shown in the following figure.



General form of Maxwell Equations for Time-varying Fields

The general form of time-varying Maxwell equations,

$$\nabla \times \bar{\mathcal{E}} = \frac{-\partial \bar{\mathcal{B}}}{\partial t}, \quad (1.1a)$$

$$\nabla \times \bar{\mathcal{H}} = \frac{\partial \bar{\mathcal{D}}}{\partial t} + \bar{\mathcal{J}}, \quad (1.1b)$$

$$\nabla \cdot \bar{\mathcal{D}} = \rho, \quad (1.1c)$$

$$\nabla \cdot \bar{\mathcal{B}} = 0. \quad (1.1d)$$

$\bar{\mathcal{E}}$ is the electric field, in volts per meter (V/m).¹

$\bar{\mathcal{H}}$ is the magnetic field, in amperes per meter (A/m).

$\bar{\mathcal{D}}$ is the electric flux density, in coulombs per meter squared (Coul/m²).

$\bar{\mathcal{B}}$ is the magnetic flux density, in webers per meter squared (Wb/m²).

$\bar{\mathcal{J}}$ is the electric current density, in amperes per meter squared (A/m²).

ρ is the electric charge density, in coulombs per meter cubed (Coul/m³).

$$\bar{\mathcal{B}} = \mu_0 \bar{\mathcal{H}},$$

$$\bar{\mathcal{D}} = \epsilon_0 \bar{\mathcal{E}},$$

We will assume cosine-based phasors in this book, so the conversion from phasor quantities to real time-varying quantities is accomplished by multiplying the phasor by $e^{j\omega t}$ and taking the real part:

$$\bar{\mathcal{E}}(x, y, z, t) = \bar{E}(x, y, z) \cos(\omega t + \phi), \quad (1.8)$$

$$\bar{\mathcal{E}}(x, y, z, t) = \text{Re}\{\bar{E}(x, y, z)e^{j\omega t}\}, \quad (1.10)$$

Assuming an $e^{j\omega t}$ time dependence, we can replace the time derivatives in (1.1a)–(1.1d) with $j\omega$. Maxwell's equations in phasor form then become

$$\nabla \times \bar{E} = -j\omega \bar{B} - \bar{M}, \quad (1.14a)$$

$$\nabla \times \bar{H} = j\omega \bar{D} + \bar{J}, \quad (1.14b)$$

$$\nabla \cdot \bar{D} = \rho, \quad (1.14c)$$

$$\nabla \cdot \bar{B} = 0. \quad (1.14d)$$

General form of Maxwell Equations for Time-varying Fields

The Helmholtz Equation

In a source-free, linear, isotropic, homogeneous region, Maxwell's curl equations in phasor form are

$$\nabla \times \bar{E} = -j\omega\mu\bar{H}, \quad (1.41a)$$

$$\nabla \times \bar{H} = j\omega\epsilon\bar{E}, \quad (1.41b)$$

and constitute two equations for the two unknowns, $\bar{E}$ and $\bar{H}$. As such, they can be solved for either $\bar{E}$ or $\bar{H}$. Taking the curl of (1.41a) and using (1.41b) gives

$$\nabla \times \nabla \times \bar{E} = -j\omega\mu\nabla \times \bar{H} = \omega^2\mu\epsilon\bar{E},$$

which is an equation for $\bar{E}$. This result can be simplified through the use of vector identity (B.14), $\nabla \times \nabla \times \bar{A} = \nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}$, which is valid for the rectangular components of an arbitrary vector $\bar{A}$. Then,

$$\nabla^2 \bar{E} + \omega^2\mu\epsilon\bar{E} = 0, \quad (1.42)$$

because $\nabla \cdot \bar{E} = 0$ in a source-free region. Equation (1.42) is the *wave equation*, or *Helmholtz equation*, for $\bar{E}$. An identical equation for $\bar{H}$ can be derived in the same manner:

$$\nabla^2 \bar{H} + \omega^2\mu\epsilon\bar{H} = 0. \quad (1.43)$$

A constant $k = \omega\sqrt{\mu\epsilon}$ is defined and called the *propagation constant* (also known as the *phase constant*, or *wave number*), of the medium; its units are 1/m.

General Solution of Maxwell Equations

GENERAL SOLUTIONS FOR TEM, TE, AND TM WAVES

In this section we will find general solutions to Maxwell's equations for the specific cases of TEM, TE, and TM wave propagation in cylindrical transmission lines or waveguides. The geometry of an arbitrary transmission line or waveguide is shown in Figure 3.1 and is characterized by conductor boundaries that are parallel to the z -axis. These structures are assumed to be uniform in shape and dimension in the z direction and infinitely long. The conductors will initially be assumed to be perfectly conducting, but attenuation can be found by the perturbation method discussed in Chapter 2.

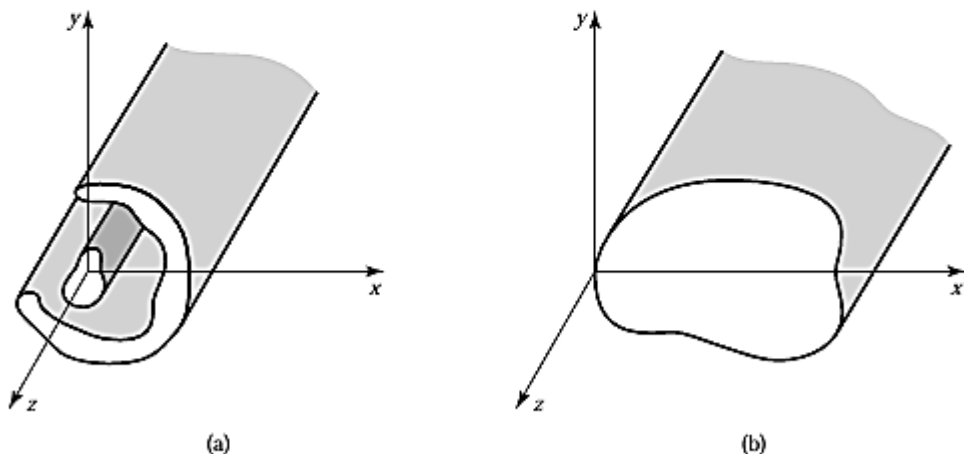


FIGURE 3.1 (a) General two-conductor transmission line and (b) closed waveguide.

We assume time-harmonic fields with an $e^{j\omega t}$ dependence and wave propagation along the z -axis. The electric and magnetic fields can then be written as

$$\vec{E}(x, y, z) = [\vec{e}(x, y) + \hat{z}e_z(x, y)]e^{-j\beta z}, \quad (3.1a)$$

$$\vec{H}(x, y, z) = [\vec{h}(x, y) + \hat{z}h_z(x, y)]e^{-j\beta z}, \quad (3.1b)$$

where $\vec{e}(x, y)$ and $\vec{h}(x, y)$ represent the transverse ($\hat{x}$, $\hat{y}$) electric and magnetic field components, and e_z and h_z are the longitudinal electric and magnetic field components. In (3.1) the wave is propagating in the $+z$ direction; $-z$ propagation can be obtained by replacing β with $-\beta$. In addition, if conductor or dielectric loss is present, the propagation constant will be complex; $j\beta$ should then be replaced with $\gamma = \alpha + j\beta$.

Assuming that the transmission line or waveguide region is source free, we can write Maxwell's equations as

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}, \quad (3.2a)$$

$$\nabla \times \vec{H} = j\omega\epsilon\vec{E}. \quad (3.2b)$$

With an $e^{-j\beta z}$ z dependence, the three components of each of these vector equations can be reduced to the following:

$$\frac{\partial E_z}{\partial y} + j\beta E_y = -j\omega\mu H_x, \quad (3.3a)$$

$$-j\beta E_x - \frac{\partial E_z}{\partial x} = -j\omega\mu H_y, \quad (3.3b)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z, \quad (3.3c)$$

$$\frac{\partial H_z}{\partial y} + j\beta H_y = j\omega\epsilon E_x, \quad (3.4a)$$

$$-j\beta H_x - \frac{\partial H_z}{\partial x} = j\omega\epsilon E_y, \quad (3.4b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\epsilon E_z. \quad (3.4c)$$

General Solution of Maxwell Equations

These six equations can be solved for the four transverse field components in terms of E_z and H_z [e.g., H_x can be derived by eliminating E_y from (3.3a) and (3.4b)] as follows:

$$H_x = \frac{j}{k_c^2} \left(\omega\epsilon \frac{\partial E_z}{\partial y} - \beta \frac{\partial H_z}{\partial x} \right), \quad (3.5a)$$

$$H_y = \frac{-j}{k_c^2} \left(\omega\epsilon \frac{\partial E_z}{\partial x} + \beta \frac{\partial H_z}{\partial y} \right), \quad (3.5b)$$

$$E_x = \frac{-j}{k_c^2} \left(\beta \frac{\partial E_z}{\partial x} + \omega\mu \frac{\partial H_z}{\partial y} \right), \quad (3.5c)$$

$$E_y = \frac{j}{k_c^2} \left(-\beta \frac{\partial E_z}{\partial y} + \omega\mu \frac{\partial H_z}{\partial x} \right), \quad (3.5d)$$

where

$$k_c^2 = k^2 - \beta^2 \quad (3.6)$$

is defined as the *cutoff wave number*; the reason for this terminology will become clear later. As in previous chapters,

$$k = \omega\sqrt{\mu\epsilon} = 2\pi/\lambda \quad (3.7)$$

is the wave number of the material filling the transmission line or waveguide region. If dielectric loss is present, ϵ can be made complex by using $\epsilon = \epsilon_0\epsilon_r(1 - j\tan\delta)$, where $\tan\delta$ is the loss tangent of the material.

Equations (3.5a)–(3.5d) are general results that can be applied to a variety of wave-guiding systems. We will now specialize these results to specific wave types.

General Solution of Maxwell Equations

TEM Waves

Transverse electromagnetic (TEM) waves are characterized by $E_z = H_z = 0$. Observe from (3.5) that if $E_z = H_z = 0$, then the transverse fields are also all zero, unless $k_c^2 = 0(k^2 = \beta^2)$, in which case we have an indeterminate result. However, we can return to (3.3)–(3.4) and apply the condition that $E_z = H_z = 0$. Then from (3.3a) and (3.4b), we can eliminate H_x to obtain

$$\beta^2 E_y = \omega^2 \mu \epsilon E_y,$$

or

$$\beta = \omega \sqrt{\mu \epsilon} = k, \quad (3.8)$$

as noted earlier. [This result can also be obtained from (3.3b) and (3.4a).] The cutoff wave number, $k_c = \sqrt{k^2 - \beta^2}$, is thus zero for TEM waves.

The Helmholtz wave equation for E_x is, from (1.42),

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) E_x = 0, \quad (3.9)$$

but for $e^{-j\beta z}$ dependence, $(\partial^2/\partial z^2) E_x = -\beta^2 E_x = -k^2 E_x$, so (3.9) reduces to

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) E_x = 0. \quad (3.10)$$

A similar result also applies to E_y , so using the form of $\bar{E}$ assumed in (3.1a), we can write

$$\nabla_t^2 \bar{e}(x, y) = 0, \quad (3.11)$$

where $\nabla_t^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the Laplacian operator in the two transverse dimensions.

The result of (3.11) shows that the transverse electric fields, $\bar{e}(x, y)$, of a TEM wave satisfy Laplace's equation. It is easy to show in the same way that the transverse magnetic fields also satisfy Laplace's equation:

$$\nabla_t^2 \bar{h}(x, y) = 0. \quad (3.12)$$

The transverse fields of a TEM wave are thus the same as the static fields that can exist between the conductors. In the electrostatic case, we know that the electric field can be expressed as the gradient of a scalar potential, $\Phi(x, y)$:

$$\bar{e}(x, y) = -\nabla_t \Phi(x, y), \quad (3.13)$$

where $\nabla_t = \hat{x}(\partial/\partial x) + \hat{y}(\partial/\partial y)$ is the transverse gradient operator in two dimensions. For the relation in (3.13) to be valid, the curl of $\bar{e}$ must vanish, and this is indeed the case here since

$$\nabla_t \times \bar{e} = -j\omega \mu h_z \hat{z} = 0.$$

Using the fact that $\nabla \cdot \bar{D} = \epsilon \nabla_t \cdot \bar{e} = 0$ with (3.13) shows that $\Phi(x, y)$ also satisfies Laplace's equation,

$$\nabla_t^2 \Phi(x, y) = 0, \quad (3.14)$$

as expected from electrostatics. The voltage between two conductors can be found as

$$V_{12} = \Phi_1 - \Phi_2 = \int_1^2 \bar{E} \cdot d\bar{\ell}, \quad (3.15)$$

where Φ_1 and Φ_2 represent the potential at conductors 1 and 2, respectively.

where Φ_1 and Φ_2 represent the potential at conductors 1 and 2, respectively. The current flow on a given conductor can be found from Ampere's law as

$$I = \oint_C \bar{H} \cdot d\bar{\ell}, \quad (3.16)$$

where C is the cross-sectional contour of the conductor.

General Solution of Maxwell Equations

A single conductor-based waveguide/ transmission line cannot provide the boundary conditions to create the required potential difference and closed loop current, necessary for TEM wave. So, TEM mode does not exist in single conductor waveguides

TEM waves can exist when two or more conductors are present. Plane waves are also examples of TEM waves since there are no field components in the direction of propagation; in this case the transmission line conductors may be considered to be two infinitely large plates separated to infinity. The above results show that a closed conductor (such as a rectangular waveguide) cannot support TEM waves since the corresponding static potential in such a region would be zero (or possibly a constant), leading to $\vec{E} = 0$.

The wave impedance of a TEM mode can be found as the ratio of the transverse electric and magnetic fields:

$$Z_{\text{TEM}} = \frac{E_x}{H_y} = \frac{\omega\mu}{\beta} = \sqrt{\frac{\mu}{\epsilon}} = \eta, \quad (3.17a)$$

where (3.4a) was used. The other pair of transverse field components, from (3.3a), gives

$$Z_{\text{TEM}} = \frac{-E_y}{H_x} = \sqrt{\frac{\mu}{\epsilon}} = \eta. \quad (3.17b)$$

TE Waves

Transverse electric (TE) waves, (also referred to as H -waves) are characterized by $E_z = 0$ and $H_z \neq 0$. Equations (3.5) then reduce to

$$H_x = \frac{-j\beta}{k_c^2} \frac{\partial H_z}{\partial x}, \quad (3.19a)$$

$$H_y = \frac{-j\beta}{k_c^2} \frac{\partial H_z}{\partial y}, \quad (3.19b)$$

$$E_x = \frac{-j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial y}, \quad (3.19c)$$

$$E_y = \frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial x}. \quad (3.19d)$$

In this case $k_c \neq 0$, and the propagation constant $\beta = \sqrt{k^2 - k_c^2}$ is generally a function of frequency and the geometry of the line or guide. To apply (3.19), one must first find H_z from the Helmholtz wave equation,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) H_z = 0, \quad (3.20)$$

which, since $H_z(x, y, z) = h_z(x, y)e^{-j\beta z}$, can be reduced to a two-dimensional wave equation for h_z :

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2 \right) h_z = 0, \quad (3.21)$$

since $k_c^2 = k^2 - \beta^2$. This equation must be solved subject to the boundary conditions of the specific guide geometry.

The TE wave impedance can be found as

$$Z_{\text{TE}} = \frac{E_x}{H_y} = \frac{-E_y}{H_x} = \frac{\omega\mu}{\beta} = \frac{k\eta}{\beta}, \quad (3.22)$$

which is seen to be frequency dependent. TE waves can be supported inside closed conductors, as well as between two or more conductors.

General Solution of Maxwell Equations

TM Waves

Transverse magnetic (TM) waves (also referred to as *E*-waves) are characterized by $E_z \neq 0$ and $H_z = 0$. Equations (3.5) then reduce to

$$H_x = \frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial y}, \quad (3.23a)$$

$$H_y = \frac{-j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial x}, \quad (3.23b)$$

$$E_x = \frac{-j\beta}{k_c^2} \frac{\partial E_z}{\partial x}, \quad (3.23c)$$

$$E_y = \frac{-j\beta}{k_c^2} \frac{\partial E_z}{\partial y}. \quad (3.23d)$$

As in the TE case, $k_c \neq 0$, and the propagation constant $\beta = \sqrt{k^2 - k_c^2}$ is a function of frequency and the geometry of the line or guide. E_z is found from the Helmholtz wave equation,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) E_z = 0, \quad (3.24)$$

which, since $E_z(x, y, z) = e_z(x, y)e^{-j\beta z}$, can be reduced to a two-dimensional wave equation for e_z :

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2 \right) e_z = 0, \quad (3.25)$$

since $k_c^2 = k^2 - \beta^2$. This equation must be solved subject to the boundary conditions of the specific guide geometry.

The TM wave impedance can be found as

$$Z_{\text{TM}} = \frac{E_x}{H_y} = \frac{-E_y}{H_x} = \frac{\beta}{\omega\epsilon} = \frac{\beta\eta}{k}, \quad (3.26)$$

which is frequency dependent. As for TE waves, TM waves can be supported inside closed conductors, as well as between two or more conductors.

Waveguides

A hollow metallic tube of uniform cross-section for transmitting electromagnetic waves by successive reflections from the inner walls of the tube is called as a **Waveguide**.

A waveguide is generally preferred in microwave communications. Waveguide is a special form of transmission line, which is a hollow metal tube. Unlike a transmission line, a waveguide has no center conductor.

The main characteristics of a Waveguide are –

- The tube wall provides distributed inductance.
- The empty space between the tube walls provide distributed capacitance.
- These are bulky and expensive.

Advantages of Waveguides

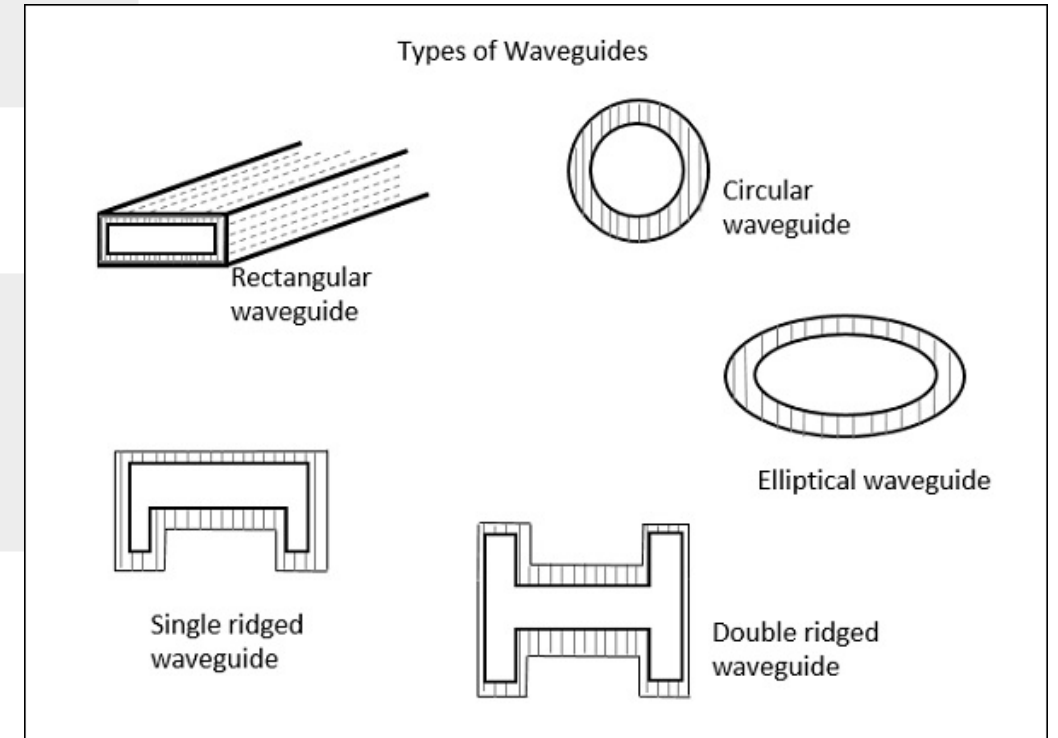
Following are few advantages of Waveguides.

- Waveguides are easy to manufacture.
- They can handle very large power (in kilo watts).
- Power loss is very negligible in waveguides.
- They offer very low loss (low value of alpha-attenuation).
- When microwave energy travels through waveguide, it experiences lower losses than a coaxial cable.

Types of Waveguides

There are five types of waveguides.

- Rectangular waveguide
- Circular waveguide
- Elliptical waveguide
- Single-ridged waveguide
- Double-ridged waveguide



Transmission Lines vs Waveguides

Transmission Lines Vs Waveguides

The main difference between a transmission line and a wave guide is –

- A **two conductor structure** that can support a TEM wave is a transmission line.
- A **one conductor structure** that can support a TE wave or a TM wave but not a TEM wave is called as a waveguide.

The following table brings out the differences between transmission lines and waveguides.

Transmission Lines	Waveguides
Supports TEM wave	Cannot support TEM wave
All frequencies can pass through	Only the frequencies that are greater than cut-off frequency can pass through
Two conductor transmission	One conductor transmission
Reflections are less	A wave travels through reflections from the walls of the waveguide
It has a characteristic impedance	It has wave impedance
Propagation of waves is according to "Circuit theory"	Propagation of waves is according to "Field theory"
It has a return conductor to earth	Return conductor is not required as the body of the waveguide acts as earth
Bandwidth is not limited	Bandwidth is limited
Waves do not disperse	Waves get dispersed

Parallel Plate Waveguides

TEM Modes

As discussed in Section 3.1, the TEM mode solution can be obtained by solving Laplace's equation, (3.14), for the electrostatic potential $\Phi(x, y)$ between the two plates. Thus,

$$\nabla_t^2 \Phi(x, y) = 0, \quad \text{for } 0 \leq x \leq W, \quad 0 \leq y \leq d. \quad (3.31)$$

Solution of Laplace's equation (3.31)

$$\bar{E}(x, y, z) = \bar{e}(x, y)e^{-jkz} = -\hat{y}\frac{V_0}{d}e^{-jkz}, \quad (3.35)$$

$$\bar{H}(x, y, z) = \bar{h}(x, y)e^{-jkz} = \frac{1}{\eta}\hat{z} \times \bar{E}(x, y, z) = \hat{x}\frac{V_0}{\eta d}e^{-jkz}, \quad (3.36)$$

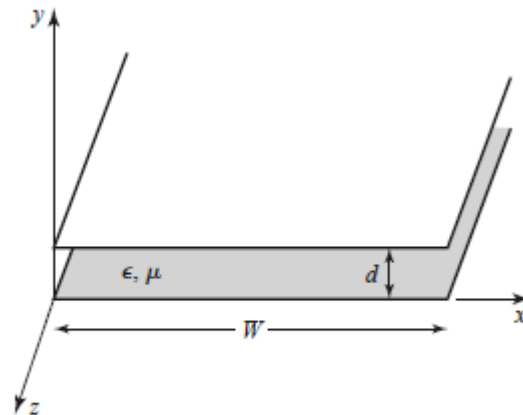
where $\eta = \sqrt{\mu/\epsilon}$ is the intrinsic impedance of the medium between the parallel plates.

the characteristic impedance is

$$Z_0 = \frac{\eta d}{W},$$

The phase velocity is also a constant:

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\epsilon}},$$



Geometry of a parallel plate waveguide.

TM Modes

As discussed in Section 3.1, TM waves are characterized by $H_z = 0$ and a nonzero E_z field that satisfies the reduced wave equation of (3.25), with $\partial/\partial x = 0$:

$$\left(\frac{\partial^2}{\partial y^2} + k_c^2\right) e_z(x, y) = 0, \quad (3.41)$$

where $k_c = \sqrt{k^2 - \beta^2}$ is the cutoff wave number, and $E_z(x, y, z) = e_z(x, y)e^{-j\beta z}$.

$$E_z(x, y, z) = A_n \sin \frac{n\pi y}{d} e^{-j\beta z}.$$

$$H_x = \frac{j\omega\epsilon}{k_c} A_n \cos \frac{n\pi y}{d} e^{-j\beta z},$$

$$E_y = \frac{-j\beta}{k_c} A_n \cos \frac{n\pi y}{d} e^{-j\beta z},$$

$$E_x = H_y = 0.$$

no propagation will occur until the frequency is such that $k > k_c$. The cutoff frequency of the TM_n mode can be found as

$$f_c = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{n}{2d\sqrt{\mu\epsilon}}. \quad (3.49)$$

$$Z_{TM} = \frac{-E_y}{H_x} = \frac{\beta}{\omega\epsilon} = \frac{\beta\eta}{k},$$

$$v_p = \frac{\omega}{\beta},$$

$$\lambda_g = \frac{2\pi}{\beta},$$

Parallel Plate Waveguides

TE Modes

TE modes, characterized by $E_z = 0$, can also propagate in a parallel plate waveguide. From (3.21), with $\partial/\partial x = 0$, H_z must satisfy the reduced wave equation,

$$\left(\frac{\partial^2}{\partial y^2} + k_c^2 \right) h_z(x, y) = 0, \quad (3.62)$$

where $k_c = \sqrt{k^2 - \beta^2}$ is the cutoff wave number and $H_z(x, y, z) = h_z(x, y)e^{-j\beta z}$.

$$H_z(x, y) = B_n \cos \frac{n\pi y}{d} e^{-j\beta z}.$$

$$E_x = \frac{j\omega\mu}{k_c} B_n \sin \frac{n\pi y}{d} e^{-j\beta z},$$

$$H_y = \frac{j\beta}{k_c} B_n \sin \frac{n\pi y}{d} e^{-j\beta z},$$

$$E_y = H_x = 0.$$

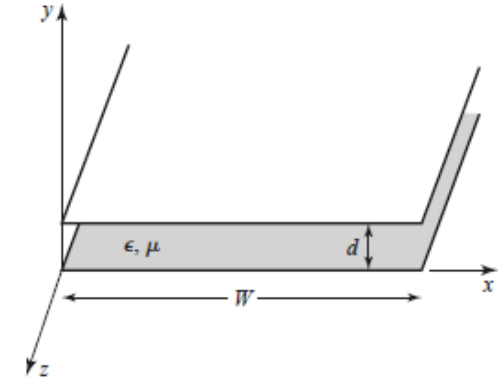
The cutoff frequency of the TE_n mode is

$$f_c = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{n}{2d\sqrt{\mu\epsilon}},$$

$$Z_{TE} = \frac{E_x}{H_y} = \frac{\omega\mu}{\beta} = \frac{k\eta}{\beta},$$

$$v_p = \frac{\omega}{\beta},$$

$$\lambda_g = \frac{2\pi}{\beta},$$



Geometry of a parallel plate waveguide.

TABLE 3.1 Summary of Results for Parallel Plate Waveguide

Quantity	TEM Mode	TM _n Mode	TE _n Mode
k	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$
k_c	0	$n\pi/d$	$n\pi/d$
β	$k = \omega\sqrt{\mu\epsilon}$	$\sqrt{k^2 - k_c^2}$	$\sqrt{k^2 - k_c^2}$
λ_c	∞	$2\pi/k_c = 2d/n$	$2\pi/k_c = 2d/n$
λ_g	$2\pi/k$	$2\pi/\beta$	$2\pi/\beta$
v_p	$\omega/k = 1/\sqrt{\mu\epsilon}$	ω/β	ω/β
α_d	$(k \tan \delta)/2$	$(k^2 \tan \delta)/2\beta$	$(k^2 \tan \delta)/2\beta$
α_c	$R_s/\eta d$	$2kR_s/\beta\eta d$	$2k_c^2 R_s/k\beta\eta d$
E_z	0	$A \sin(n\pi y/d)e^{-j\beta z}$	0
H_z	0	0	$B \cos(n\pi y/d)e^{-j\beta z}$
E_x	0	0	$(j\omega\mu/k_c)B \sin(n\pi y/d)e^{-j\beta z}$
E_y	$(-V_0/d)e^{-j\beta z}$	$(-j\beta/k_c)A \cos(n\pi y/d)e^{-j\beta z}$	0
H_x	$(V_0/\eta d)e^{-j\beta z}$	$(j\omega\epsilon/k_c)A \cos(n\pi y/d)e^{-j\beta z}$	0
H_y	0	0	$(j\beta/k_c)B_n \sin(n\pi y/d)e^{-j\beta z}$
Z	$Z_{TEM} = \eta d/W$	$Z_{TM} = \beta\eta/k$	$Z_{TE} = k\eta/\beta$

Rectangular Waveguides

TE Modes

The geometry of a rectangular waveguide is shown in Figure 3.7, where it is assumed that the guide is filled with a material of permittivity ϵ and permeability μ . It is standard convention to have the longest side of the waveguide along the x -axis, so that $a > b$.

TE waveguide modes are characterized by fields with $E_z = 0$, while H_z must satisfy the reduced wave equation of (3.21):

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2 \right) h_z(x, y) = 0, \quad (3.73)$$

$$H_z(x, y, z) = A_{mn} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z},$$

The transverse field components of the TE_{mn} mode can be found using (3.19) and (3.81):

$$E_x = \frac{j\omega\mu n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.82a)$$

$$E_y = \frac{-j\omega\mu m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.82b)$$

$$H_x = \frac{j\beta m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.82c)$$

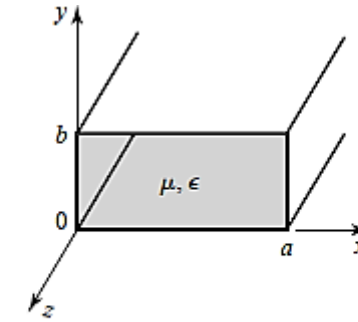
$$H_y = \frac{j\beta n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}. \quad (3.82d)$$

Each mode (each combination of m and n) has a cutoff frequency $f_{c_{mn}}$ given by

$$f_{c_{mn}} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}. \quad (3.84)$$

The mode with the lowest cutoff frequency is called the dominant mode; because we have assumed $a > b$, the lowest cutoff frequency occurs for the TE_{10} ($m = 1, n = 0$) mode:

$$f_{c_{10}} = \frac{1}{2a\sqrt{\mu\epsilon}}. \quad (3.85)$$



Geometry of a rectangular waveguide.

$$f_{c_{mn}} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}}$$

At a given operating frequency f only those modes having $f > f_c$ will propagate; modes with $f < f_c$ will lead to an imaginary β (or real α), meaning that all field components will decay exponentially away from the source of excitation. Such modes are referred to as cutoff modes, or evanescent modes. If more than one mode is propagating, the waveguide is said to be *overmoded*.

Rectangular Waveguides

TM Modes

TM modes are characterized by fields with $H_z = 0$, while E_z must satisfy the reduced wave equation (3.25):

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2 \right) e_z(x, y) = 0, \quad (3.97)$$

$$E_z(x, y, z) = B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.100)$$

where B_{mn} is an arbitrary amplitude constant.

The transverse field components for the TM_{mn} mode can be computed from (3.23) and (3.100) as

$$E_x = \frac{-j\beta m\pi}{ak_c^2} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101a)$$

$$E_y = \frac{-j\beta n\pi}{bk_c^2} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101b)$$

$$H_x = \frac{j\omega\epsilon n\pi}{bk_c^2} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101c)$$

$$H_y = \frac{-j\omega\epsilon m\pi}{ak_c^2} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}. \quad (3.101d)$$

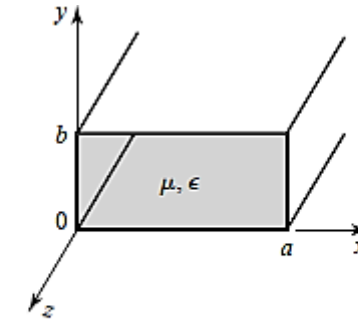
Each mode (each combination of m and n) has a cutoff frequency $f_{c_{mn}}$ given by

$$f_{c_{mn}} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}. \quad (3.84)$$

The mode with the lowest cutoff frequency is called the dominant mode;

Observe that the field expressions for $\vec{E}$ and $\vec{H}$ in (3.101) are identically zero if either m or n is zero. Thus there is no TM_{00} , TM_{01} , or TM_{10} mode, and the lowest order TM mode to propagate (lowest f_c) is the TM_{11} mode, having a cutoff frequency of

$$f_{c_{11}} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2}, \quad (3.103)$$



Geometry of a rectangular waveguide.

$$f_{c_{mn}} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}}$$

At a given operating frequency f only those modes having $f > f_c$ will propagate; modes with $f < f_c$ will lead to an imaginary β (or real α), meaning that all field components will decay exponentially away from the source of excitation. Such modes are referred to as *cutoff modes*, or *evanescent modes*. If more than one mode is propagating, the waveguide is said to be *overmoded*.

Rectangular Waveguides

EXAMPLE 3.1 CHARACTERISTICS OF A RECTANGULAR WAVEGUIDE

Consider a length of Teflon-filled, copper K-band rectangular waveguide having dimensions $a = 1.07$ cm and $b = 0.43$ cm. Find the cutoff frequencies of the first five propagating modes. If the operating frequency is 15 GHz, find the attenuation due to dielectric and conductor losses.

Solution

From Appendix G, for Teflon, $\epsilon_r = 2.08$ and $\tan \delta = 0.0004$. From (3.84) the cutoff frequencies are given by

$$f_{c_{mn}} = \frac{c}{2\pi\sqrt{\epsilon_r}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}.$$

Computing f_c for the first few values of m and n gives the following results:

Mode	m	n	f_c (GHz)
TE	1	0	9.72
TE	2	0	19.44
TE	0	1	24.19
TE, TM	1	1	26.07
TE, TM	2	1	31.03

Thus the TE₁₀, TE₂₀, TE₀₁, TE₁₁, and TM₁₁ modes will be the first five modes to propagate.

At 15 GHz, $k = 453.1 \text{ m}^{-1}$, and the propagation constant for the TE₁₀ mode is

$$\beta = \sqrt{\left(\frac{2\pi f \sqrt{\epsilon_r}}{c}\right)^2 - \left(\frac{\pi}{a}\right)^2} = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2} = 345.1 \text{ m}^{-1}.$$

From (3.29), the attenuation due to dielectric loss is

$$\alpha_d = \frac{k^2 \tan \delta}{2\beta} = 0.119 \text{ Np/m} = 1.03 \text{ dB/m}.$$

The surface resistivity of the copper walls is ($\sigma = 5.8 \times 10^7 \text{ S/m}$)

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}} = 0.032 \Omega,$$

and the attenuation due to conductor loss, from (3.96), is

$$\alpha_c = \frac{R_s}{a^3 b \beta k \eta} (2b\pi^2 + a^3 k^2) = 0.050 \text{ Np/m} = 0.434 \text{ dB/m}.$$

TABLE 3.2 Summary of Results for Rectangular Waveguide

Quantity	TE _{mn} Mode	TM _{mn} Mode
k	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$
k_c	$\sqrt{(m\pi/a)^2 + (n\pi/b)^2}$	$\sqrt{(m\pi/a)^2 + (n\pi/b)^2}$
β	$\sqrt{k^2 - k_c^2}$	$\sqrt{k^2 - k_c^2}$
λ_c	$\frac{2\pi}{k_c}$	$\frac{2\pi}{k_c}$
λ_g	$\frac{2\pi}{\beta}$	$\frac{2\pi}{\beta}$
v_p	$\frac{\omega}{\beta}$	$\frac{\omega}{\beta}$
α_d	$\frac{k^2 \tan \delta}{2\beta}$	$\frac{k^2 \tan \delta}{2\beta}$
E_z	0	$B \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
H_z	$A \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	0
E_x	$\frac{j\omega\mu n\pi}{k_c^2 b} A \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\beta m\pi}{k_c^2 a} B \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
E_y	$\frac{-j\omega\mu m\pi}{k_c^2 a} A \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\beta n\pi}{k_c^2 b} B \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$
H_x	$\frac{j\beta m\pi}{k_c^2 a} A \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{j\omega\epsilon n\pi}{k_c^2 b} B \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$
H_y	$\frac{j\beta n\pi}{k_c^2 b} A \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$	$\frac{-j\omega\epsilon m\pi}{k_c^2 a} B \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$
Z	$Z_{\text{TE}} = \frac{k\eta}{\beta}$	$Z_{\text{TM}} = \frac{\beta\eta}{k}$

Note the field expressions for TM mode

$$E_z(x, y, z) = B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.100)$$

$$E_x = \frac{-j\beta m\pi}{ak_c^2} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101a)$$

$$E_y = \frac{-j\beta n\pi}{bk_c^2} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101b)$$

$$H_x = \frac{j\omega\epsilon n\pi}{bk_c^2} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}, \quad (3.101c)$$

$$H_y = \frac{-j\omega\epsilon m\pi}{ak_c^2} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}. \quad (3.101d)$$

Observe that the field expressions for $\vec{E}$ and $\vec{H}$ in (3.101) are identically zero if either m or n is zero. Thus there is no TM_{00} , TM_{01} , or TM_{10} mode, and the lowest order TM mode to propagate (lowest f_c) is the TM_{11} mode, having a cutoff frequency of

$$f_{c11} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2}, \quad (3.103)$$

Strip Lines

In practice stripline is usually constructed by etching the center conductor on a grounded dielectric substrate of thickness $b/2$ and then covering with another grounded substrate.

Variations of the basic geometry of Figure 3.22a include stripline with differing dielectric substrate thicknesses (asymmetric stripline) or different dielectric constants (inhomogeneous stripline).

Air dielectric is sometimes used when it is necessary to minimize loss.

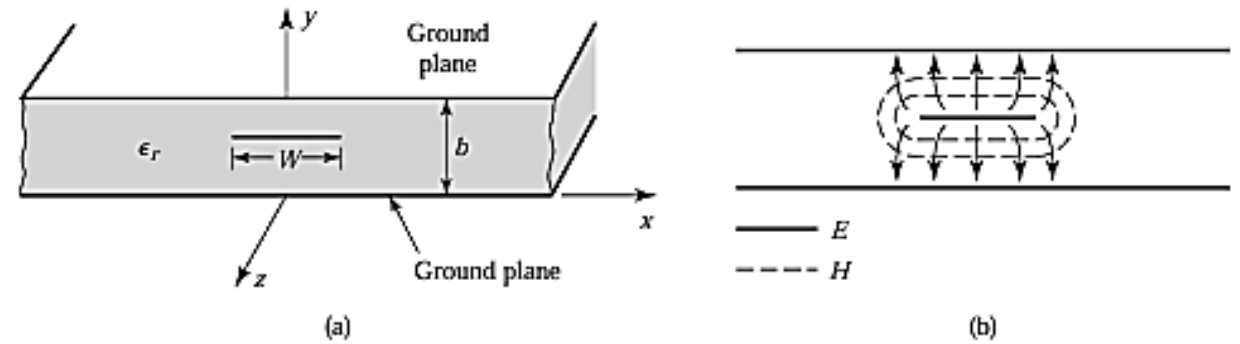
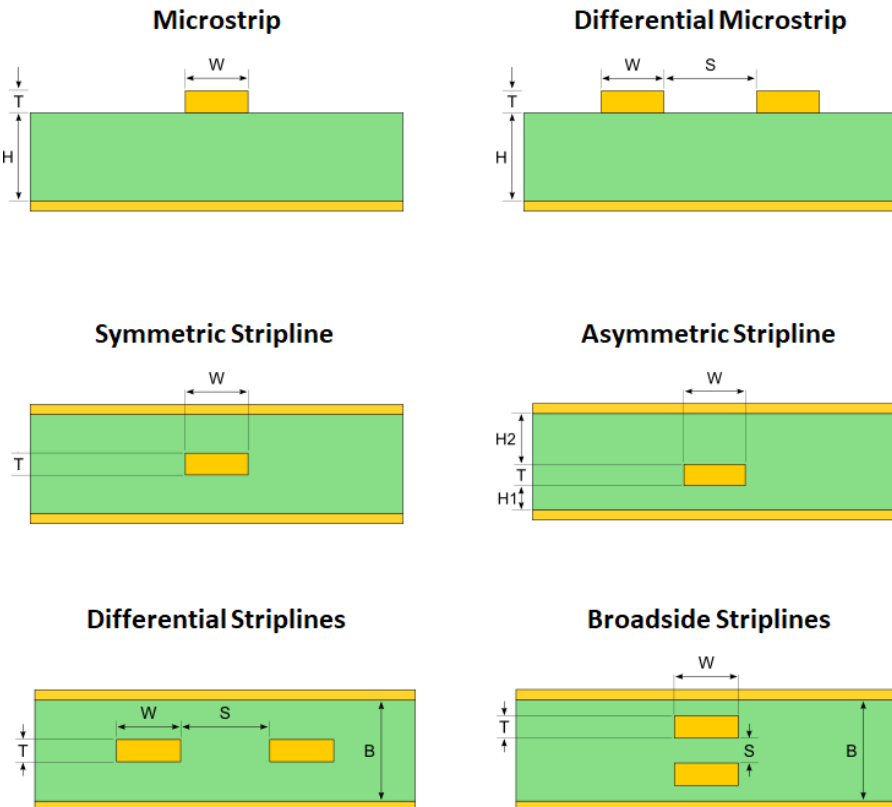


FIGURE 3.22 Stripline transmission line. (a) Geometry. (b) Electric and magnetic field lines.

Because stripline has two conductors and a homogeneous dielectric, it supports a TEM wave, and this is the usual mode of operation

Stripline can also support higher order waveguide modes. These can usually be avoided in practice by restricting both the ground plane spacing and the sidewall width to less than $\lambda_d/2$.



Strip Lines

An example of a stripline circuit is shown in Figure 3.23.

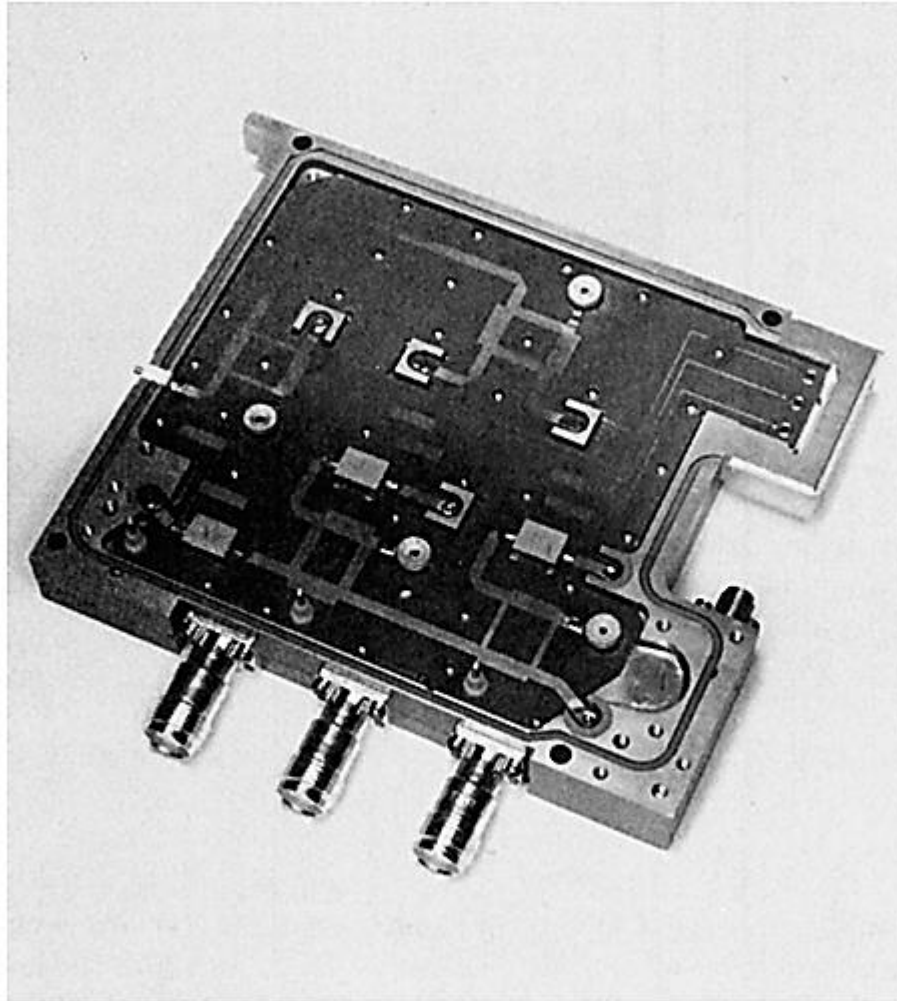


FIGURE 3.23 Photograph of a stripline circuit assembly (cover removed), showing four quadrature hybrids, open-circuit tuning stubs, and coaxial transitions.

Microstrip Lines

Microstrip line is one of the most popular types of planar transmission lines primarily because it can be fabricated by photolithographic processes and is easily miniaturized and integrated with both passive and active microwave devices.

The presence of the dielectric, particularly the fact that the dielectric does not fill the region above the strip ($y > d$), complicates the behavior and analysis of microstrip line.

Unlike stripline, where all the fields are contained within a homogeneous dielectric region, microstrip has some (usually most) of its field lines in the dielectric region between the strip conductor and the ground plane and some fraction in the air region above the substrate. For this reason microstrip line cannot support a pure TEM wave since the phase velocity of TEM fields in the dielectric region would be $c/\sqrt{\epsilon_r}$, while the phase velocity of TEM fields in the air region would be c , so a phase-matching condition at the dielectric-air interface would be impossible to enforce.

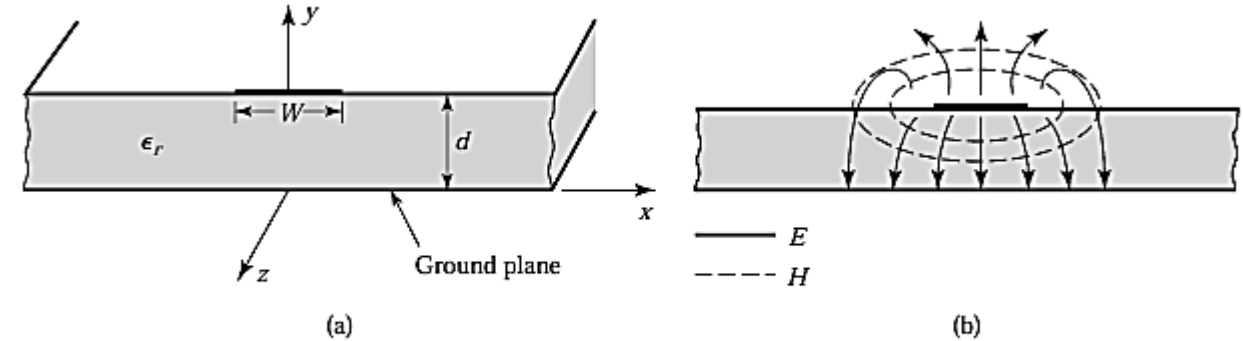
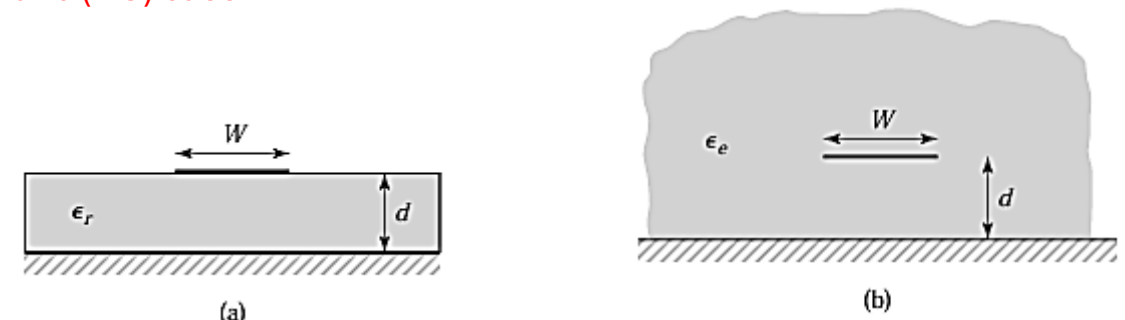


FIGURE 3.25 Microstrip transmission line. (a) Geometry. (b) Electric and magnetic field lines.

In actuality, the exact fields of a microstrip line constitute a hybrid TM-TE wave and require more advanced analysis techniques than we are prepared to deal with here. In most practical applications, however, the dielectric substrate is electrically very thin ($d \ll \lambda$), and so the fields are **quasi-TEM**. In other words, the fields are essentially the same as those of the static (DC) case.



Equivalent geometry of a quasi-TEM microstrip line. (a) Original geometry. (b) Equivalent geometry, where the dielectric substrate of relative permittivity ϵ_r is replaced with a homogeneous medium of effective relative permittivity ϵ_e .

Microstrip Lines

$$v_p = \frac{c}{\sqrt{\epsilon_e}},$$

$$\beta = k_0 \sqrt{\epsilon_e},$$

where ϵ_e is the *effective dielectric constant* of the microstrip line. Because some of the field lines are in the dielectric region and some are in air, the effective dielectric constant satisfies the relation

$$1 < \epsilon_e < \epsilon_r$$

and depends on the substrate dielectric constant, the substrate thickness, the conductor width, and the frequency.

The effective dielectric constant of a microstrip line is given approximately by

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12d/W}}. \quad (3.195)$$

The effective dielectric constant can be interpreted as the dielectric constant of a homogeneous medium that equivalently replaces the air and dielectric regions of the microstrip line, as shown in Figure 3.26. The phase velocity and propagation constant are then given by (3.193) and (3.194).

Given the dimensions of the microstrip line, the characteristic impedance can be calculated as

$$Z_0 = \begin{cases} \frac{60}{\sqrt{\epsilon_e}} \ln \left(\frac{8d}{W} + \frac{W}{4d} \right) & \text{for } W/d \leq 1 \\ \frac{120\pi}{\sqrt{\epsilon_e} [W/d + 1.393 + 0.667 \ln(W/d + 1.444)]} & \text{for } W/d \geq 1. \end{cases} \quad (3.196)$$

For a given characteristic impedance Z_0 and dielectric constant ϵ_r , the W/d ratio can be found as

$$\frac{W}{d} = \begin{cases} \frac{8e^A}{e^{2A} - 2} & \text{for } W/d < 2 \\ \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left\{ \ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right\} \right] & \text{for } W/d > 2, \end{cases} \quad (3.197)$$

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right)$$

$$B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}.$$

Microstrip Lines

Considering a microstrip line as a quasi-TEM line, we can determine the attenuation due to dielectric loss as

$$\alpha_d = \frac{k_0 \epsilon_r (\epsilon_e - 1) \tan \delta}{2 \sqrt{\epsilon_e} (\epsilon_r - 1)} \text{ Np/m}, \quad (3.198)$$

where $\tan \delta$ is the loss tangent of the dielectric. This result is derived from (3.30) by multiplying by a “filling factor,”

$$\frac{\epsilon_r (\epsilon_e - 1)}{\epsilon_e (\epsilon_r - 1)},$$

which accounts for the fact that the fields around the microstrip line are partly in air (lossless) and partly in the dielectric (lossy). The attenuation due to conductor loss is given approximately by [8]

$$\alpha_c = \frac{R_s}{Z_0 W} \text{ Np/m}, \quad (3.199)$$

where $R_s = \sqrt{\omega \mu_0 / 2 \sigma}$ is the surface resistivity of the conductor. For most microstrip substrates, conductor loss is more significant than dielectric loss; exceptions may occur, however, with some semiconductor substrates.

EXAMPLE 3.7 MICROSTRIP LINE DESIGN

Design a microstrip line on a 0.5 mm alumina substrate ($\epsilon_r = 9.9$, $\tan \delta = 0.001$) for a 50Ω characteristic impedance. Find the length of this line required to produce a phase delay of 270° at 10 GHz, and compute the total loss on this line, assuming copper conductors. Compare the results obtained from the approximate formulas of (3.195)–(3.199) with those from a microwave CAD package.

Solution

First find W/d for $Z_0 = 50 \Omega$, and initially guess that $W/d < 2$. From (3.197),

$$A = 2.142, \quad W/d = 0.9654.$$

So the condition that $W/d < 2$ is satisfied; otherwise we would use the expression for $W/d > 2$. Then the required line width is $W = 0.9654d = 0.483$ mm. From (3.195) the effective dielectric constant is $\epsilon_e = 6.665$. The line length, ℓ , for a 270° phase shift is found as

$$\begin{aligned} \phi &= 270^\circ = \beta \ell = \sqrt{\epsilon_e} k_0 \ell, \\ k_0 &= \frac{2\pi f}{c} = 209.4 \text{ m}^{-1}, \\ \ell &= \frac{270^\circ (\pi/180^\circ)}{\sqrt{\epsilon_e} k_0} = 8.72 \text{ mm}. \end{aligned}$$

Attenuation due to dielectric loss is found from (3.198) as $\alpha_d = 0.255 \text{ Np/m} = 0.022 \text{ dB/cm}$. The surface resistivity for copper at 10 GHz is 0.026Ω , and the attenuation due to conductor loss is, from (3.199), $\alpha_c = 0.0108 \text{ Np/cm} = 0.094 \text{ dB/cm}$. The total loss on the line is then 0.101 dB.

A commercial microwave CAD package gives the following results: $W = 0.478$ mm, $\epsilon_e = 6.83$, $\ell = 8.61$ mm, $\alpha_d = 0.022 \text{ dB/cm}$, and $\alpha_c = 0.054 \text{ dB/cm}$. The approximate formulas give results that are within a few percent of the CAD data for linewidth, effective dielectric constant, line length, and dielectric attenuation. The greatest discrepancy occurs for the attenuation constant for conductor loss. ■